Final Report for Wildfire Risk Modeling Exercise:

Module A – Town of Forests

Chenzhi Ma, Majid Bavandpour, Hamed Ebrahimian*

Wildfire Science and Engineering Lab (https://www.wildfirehub.org/)
Department of Civil and Environmental Engineering, University of Nevada, Reno, USA
**PI, Email: hebrahimian@unr.edu*

1. Workflow Description

We utilized a three-step modular framework to assess parcel-level wildfire risk. In this context, risk is defined as the spatial probability distribution of the damage and loss metrics across the wildland–urban interface (WUI) domain. Risk is calculated by quantifying and propagating uncertainties in key wildfire driver (or input) parameters including, weather, fuel, and ignition through models that simulate fire spread and structural-level damage. These uncertainties are informed by historical data for the Town of Forests, and the resulting risk estimates are, therefore, conditioned on historical patterns, making them most applicable to near-term scenarios (e.g., next fire seasons) rather than long-term climatic projections.

The three-step framework is structured as follows. In the first step, spatial burn probability is computed across the wildland and WUI landscape, representing the average annual probability that the fire reaches each location. In the second step, a machine learning–based model is employed to estimate the conditional probability of structural damage within the WUI, given fire exposure at a given location. In the third step, the outputs of the first two components are integrated to derive quantitative risk metrics, including the annual probability of parcel-level damage, the expected annual loss (EAL) per parcel, and the aggregated expected annual loss at the community scale.

The work begins with burn probability (BP) estimation, as shown in Figure 1. Burn probability is defined as the probability of annual burn across the landscape. The BP module quantifies burn probability by treating wildfire behavior as the outcome of uncertain environmental and ignition conditions, rather than a single deterministic scenario. It systematically integrates uncertainties in key drivers—such as wind speed, wind direction, fuel load and fuel moisture, and ignition—through a fire simulation framework, here FARSITE. To ensure computational tractability while preserving fidelity to the underlying uncertainty space, the approach adopts an efficient deterministic sampling strategy in lieu of fully stochastic (e.g., Monte Carlo) simulation. This enables structured exploration of the input space and quantification of how variations in drivers influence fire behavior metrics, such as rate of spread and flame length. The resulting ensemble of simulations is spatially aggregated to estimate the probability that each location burns, thereby constructing a probabilistic representation of fire occurrence across the domain. The BP estimates provide a spatially explicit and statistically grounded characterization of wildfire exposure,

capturing both the inherent variability in environmental conditions and the nonlinear dynamics of fire spread. As such, they form the foundational layer for subsequent risk quantification by linking uncertain drivers to the likelihood of fire impact at the parcel and community scales.

The second step performs building-level damage prediction. While the first step characterizes where fire is likely to occur across the landscape, this module evaluates how individual buildings are expected to respond once exposed to wildfire. In this step, we develop a CATBOOST-based machine learning model to predict building damage probability. The model inputs integrate building attributes with physics-guided indicators of key exposure pathways, including direct flame contact, radiative heating from nearby buildings and vegetation, and embers from upwind buildings and vegetations. The model output is the structure/parcel-specific damage probability. In this way, the second step translates spatial wildfire hazard into actionable estimates of structural vulnerability, enabling risk quantification at both parcel and community scales.

The final stage aggregates risk. For each building, the burning probability $(p(bp))$ from the first step is combined with the structural/parcel-level damage probability $(p(d|bp))$ from the second step. The two values are then used to estimate expected annual loss (EAL) by multiplying with the average building loss and the building area A. Here, EAL represents the expected annual loss for a specific WUI building, accounting for both the likelihood of burning and the likelihood of damage once exposed to fire.

Overall, the workflow is designed as a modular three-step pipeline: wildfire spread uncertainty is translated into spatial burn likelihood; burn likelihood is linked to building-specific vulnerability; and the combined outputs are aggregated into EAL. This structure makes the workflow suitable not only for the present exercise, but also for future scenario analysis, mitigation testing, and decision support.

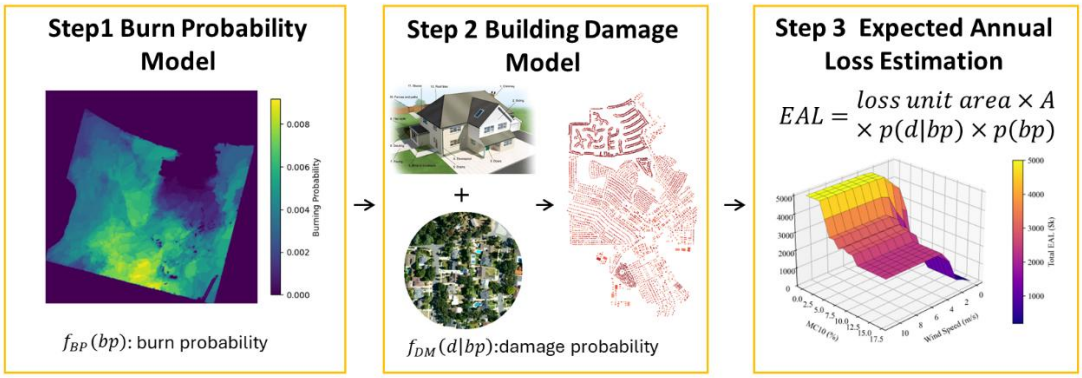


Figure 1 Workflow Diagram

2. Model Architecture and Explanation

The overall model architecture consists of three connected components: a burn-probability model, a building-damage model, and a loss estimation model. The first captures spatial probability of fire occurrence, the second captures structure damage conditional on fire occurrence, and the third captures individual building and community level expected annual loss under WUI fire.

2.1 Burn Probability Model

Burn probability quantification is performed through a two-stage process consisting of (i) ignition probability estimation, which is referred to as hazard analysis and (ii) characterization and propagation of environmental uncertainties through a fire propagation simulator, which is referred to as the system analysis component. Ignition occurrence is modeled as a stochastic point process under a Poisson assumption, where the number of ignition events over a specified spatial and temporal domain follows a Poisson distribution with rate parameter (λ) inferred from historical fire occurrence data. This rate parameter is allowed to vary spatially to reflect heterogeneity in ignition occurrence associated with factors such as fuel conditions, human activity, and historical fire regimes. In this formulation, ignition likelihood is not prescribed heuristically but is statistically grounded in observed data, enabling consistent estimation of both the expected frequency and spatial variability of ignitions. The resulting spatially distributed Poisson rates are subsequently used to weight ignition scenarios, such that regions with higher historical ignition frequency contribute proportionally more to the overall burn probability outcome.

Uncertainty in wind speed, wind direction, fuel load, and fuel moisture content (FMC) is represented probabilistically and propagated through the wildfire simulation model within the system analysis stage. Wind speed is modeled as a lognormal random variable to capture its non-negative and skewed characteristics, while wind direction is represented using a mixed Gaussian distribution to account for multimodal prevailing wind patterns. FMC is treated as an uncertain input reflecting variability in fuel conditions and is incorporated through appropriate probabilistic or discretized representations. These uncertain inputs, along with their dependencies, are propagated using the Unscented Transform, which deterministically selects a set of sampling (or sigma) points to approximate the input distribution. Model evaluations at these sigma points are then recombined to approximate the distribution of fire behavior outputs. This approach enables efficient and accurate propagation of nonlinear effects, ensuring that the resulting burn probability integrates both ignition likelihood and the full range of plausible environmental conditions. Further technical details are presented in a journal publication currently under review and can be shared upon request.

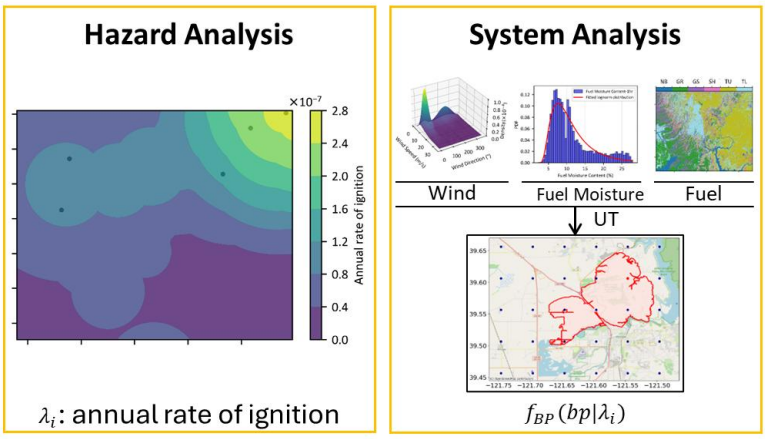


Figure 2 Burn probability quantification workflow diagram

2.2 Building Damage Model

The building-damage module is an interpretable machine-learning model that predicts damage probability for individual structures (Figure 3). Building damage in wildfire events is governed by multiple interacting physical mechanisms, including direct flame contact, thermal radiation from nearby fuels and structures, and ember attack, in addition to intrinsic building characteristics and local environmental conditions. While these processes can be described using high-fidelity physics-based models, such approaches are not computationally tractable or data-feasible at community scale.

To address this limitation, we develop an interpretable CatBoost-based model that enables efficient, structure/parcel-level damage prediction while preserving physical relevance. The model receives building features (including structural category, roof construction type, fence characteristics, year built, siding and window characteristics, eave and vent characteristics, and deck and patio characteristics), flame-contact potential (represented by proximity and configuration of adjacent structures and vegetation, i.e., within 3 m, radiative exposure (represented by geometry-based descriptors of nearby buildings and trees within 20 m), and ember exposure (represented through building and vegetative fuel in upwind direction within 60 m by 1000 m corridor). The model output is the probability of building damage given fire ($p(d|bp)$).

The model is trained using historic records of building damages in wildland and WUI fires. A comprehensive training dataset is constructed by integrating CAL FIRE Damage Inspection (DINS), weather information, neighborhood building information, and vegetation-related information for all available wildfire events from 2018. A central design principle is the incorporation of physics-guided features rather than relying solely on empirical/data-based predictors. Specifically, the feature set includes surrogate indicators representing dominant exposure pathways, direct flame contact, radiative heating, and ember transport, derived from spatial configuration, fuel characteristics, and wind conditions. This hybrid formulation enhances model interpretability, supports mechanistic consistency, and improves the utility of predictions for mitigation planning and risk-informed decision-making. This model is currently being prepared for a journal publication and further technical details can be provided upon request.

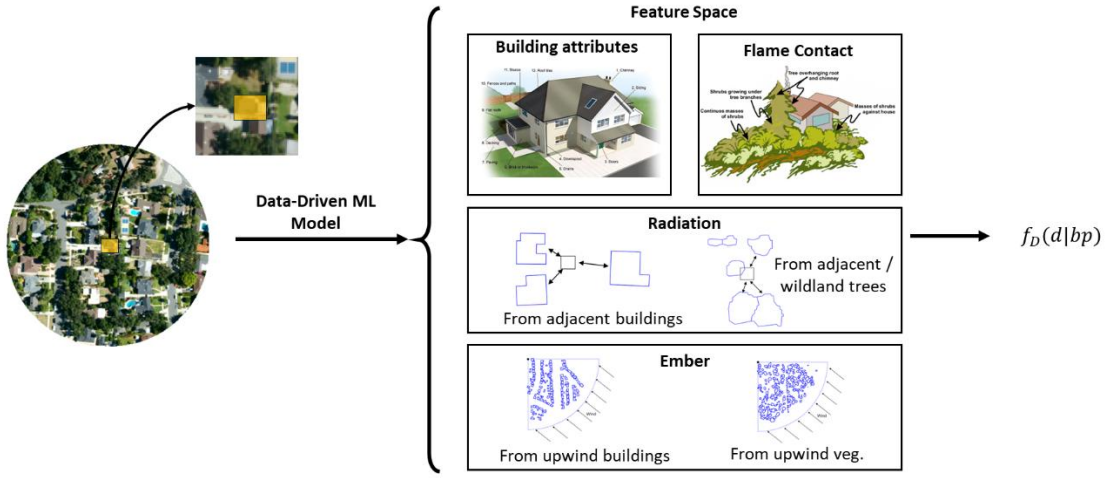


Figure 3 Diagram for building damage model

2.3 Expected Annual Loss Estimation

This part combines the output of the two models explained in previous subsections. The burn probability model estimates where fire is likely to occur. The building damage model estimates which building is likely to be damaged given fire. The burn probability and building damage probability are integrated here to find EAL of both individual building and the WUI community, as shown in the following equation.

$$EAL = Loss\ unit\ area \times A \times p(d|bp) \times p(bp) \tag{1}$$

In this equation, Loss unit area is the approximate direct monetary loss per unit area of the building (here assumed to be \$200/sqft), A is the area of the affected building/structure, and $p(d|bp)$ and $p(bp)$ are the outcomes of the models explained. It should be noted that Eq. (1) will result in expected annual loss (average loss per year) since the burn probability is characterized as the annual burn probability.

It should be noted that other metrics aside from direct monetary loss can be considered depending on the stakeholder need and preference (e.g., civil infrastructure affected, roads affected, economic-based measures, number of people affected, social-vulnerability-based loss metrics, etc.). For the purpose of this exercise, we present the EAL as it is both simple and actionable.

3. Inputs Used

3.1 Variable Inputs for Burn Probability Model

The landscape characterization in this study is based on the input datasets provided as part of the exercise, which define the spatial distribution of fuels and topographic attributes used in the wildfire simulations. Historical ignition information is obtained from the Fire Occurrence Database (FOD), which provides georeferenced records of past fire events and is used to estimate spatially varying ignition rates under the Poisson framework. Meteorological conditions are informed by observations from the Remote Automated

Weather Stations (RAWS), specifically the BARON RAWS Station, from which historical wind and fuel moisture–related variables are derived to construct the probabilistic representations of environmental inputs.

3.2 Variable Inputs for Building Damage Model

The building-damage model in this study is applied using the input dataset provided in the exercise. Building attributes, including roof characteristics, fence characteristics, year built, siding characteristics, vent-screen characteristics, eave characteristics, and deck-related characteristics, are reformatted to be consistent with the feature structure used to train the machine-learning model. Weather variables, including daily average wind speed (m/s), daily average wind direction, and 10-hour fuel moisture (%), are likewise taken directly from the exercise dataset. In addition, proxy variables for the three primary pathways of structure ignition and damage are computed building geometry provided by the exercise together with the Meta canopy height product. To ensure consistency with the trained model, the original tree-height data are not used and are instead replaced by the Meta canopy height product.

3.3 Variable Inputs for Expected Annual Loss Estimation

To estimate the building EAL given burn probability and building damage probability, loss per unit area is needed. In this exercise, we assume that all the buildings have same loss per unit area when being damaged. The unit loss value selected here is $200 per square foot.

4. Model Output

4.1 Burn Probability Model

Figure 4 presents the spatial distribution of annual burn probability at each grid size (2m × 2m) over the study domain used in the exercise. The results indicate a clear spatial gradient in burn probability across the region, with notably higher values observed in the southern portion of the city. This pattern is primarily driven by the surrounding landscape structure and fuel continuity, which support more favorable conditions for fire spread. In contrast, the northern part of the city exhibits comparatively lower burn probabilities due to its proximity to Lake Tahoe, where the presence of the lake acts as a natural barrier that limits fire propagation and reduces the likelihood of sustained burning.

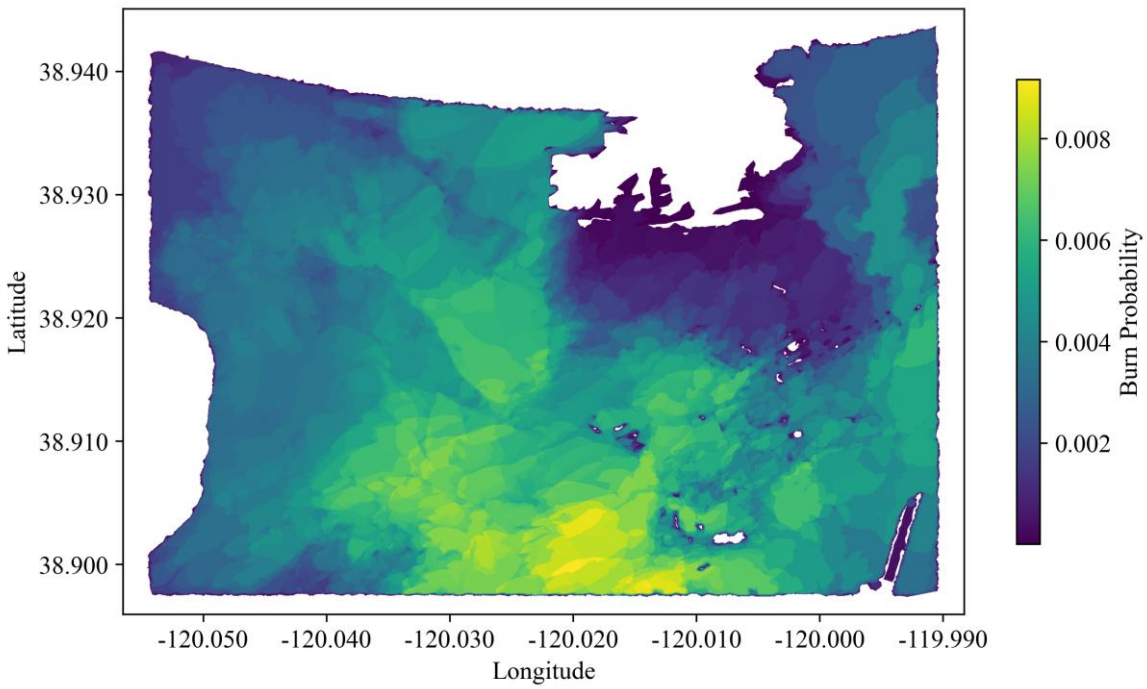
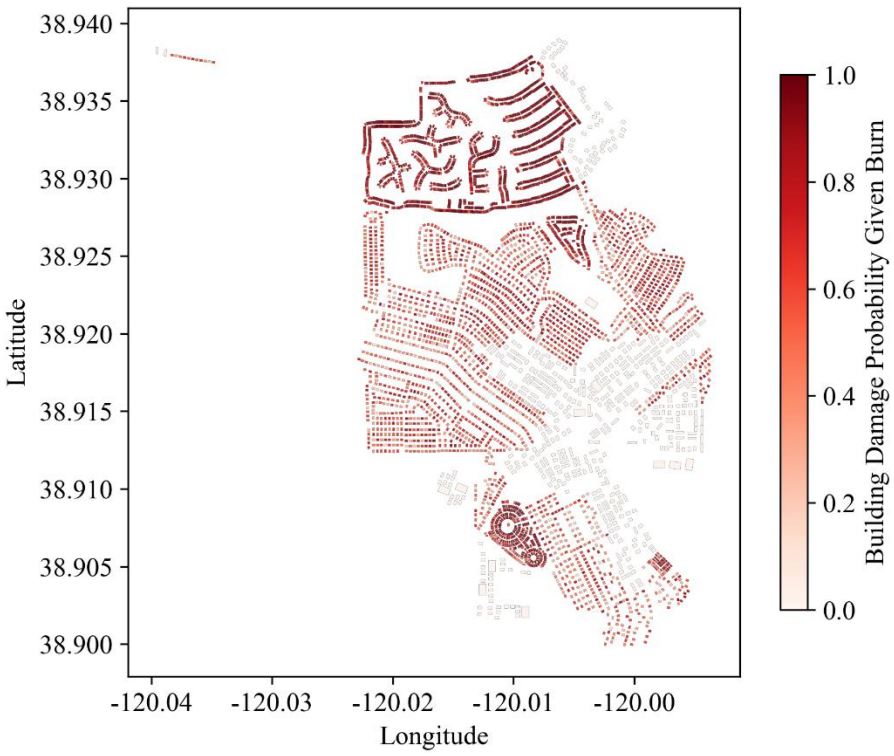


Figure 4 Annual burn probability of town of forests.

4.2 Building Damage Model

The building damage model predicts the probability of damage for each individual building in the Town of Forests scenario. Figure 5 (a) shows the spatial distribution of building-level damage probability given burn/fire under the weather conditions specified in the exercise (wind speed of 10 m/s and MC10 of 4.9%). Each building is assigned a continuous probability value between 0 and 1. Higher values indicate that the building is more likely to experience wildfire-induced damage, given its structural characteristics, local exposure, surrounding fuels, and environmental conditions.

To convert the continuous damage-probability output into a binary damaged/undamaged classification, a probability threshold is applied. In this exercise, a threshold of 0.5 is used, and 3,186 buildings are predicted as damaged. Buildings with predicted damage probability greater than 0.5 are classified as damaged, while buildings with predicted damage probability less than or equal to 0.5 are classified as undamaged. Figure 5 (b) shows the resulting spatial distribution of classified damaged and undamaged buildings. Here we assume commercial buildings and industrial buildings are non-combustible under WUI fire.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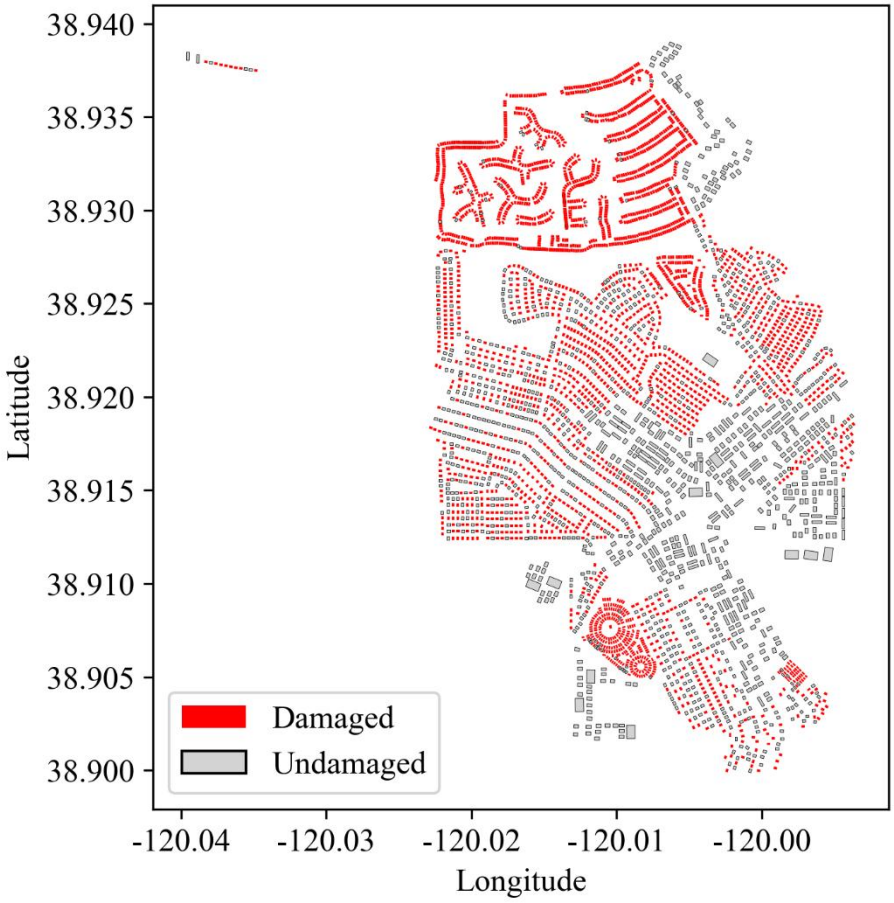
Building Damage Probability Given Burn

(a)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-120.02
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Longitude
Damaged
Undamaged

(b)

Figure 5 Building damage prediction under the exercise weather condition: (a) building damage probability and (b) classified damaged and undamaged buildings using a probability threshold of 0.5.

4.3 Expected Annual Loss Estimation

By combining the outputs of the burn probability model and the building damage model, the annual probability of building damage can be estimated, as shown in Figure 6. Unlike the probability presented in Figure 5, which represents the conditional damage probability given that a wildfire occurs in the surrounding landscape, the annual building damage probability presented here explicitly incorporates the burn probability. By further multiplying this annual damage probability by the unit loss value and building area, the EAL can be calculated for each individual building as well as for the entire community under WUI fire. For this exercise, under the default weather conditions, the building-level EAL is shown in Figure 7, and the total community-level EAL is estimated to be approximately $3,230 thousand, assuming a unit loss of $200 per square foot.

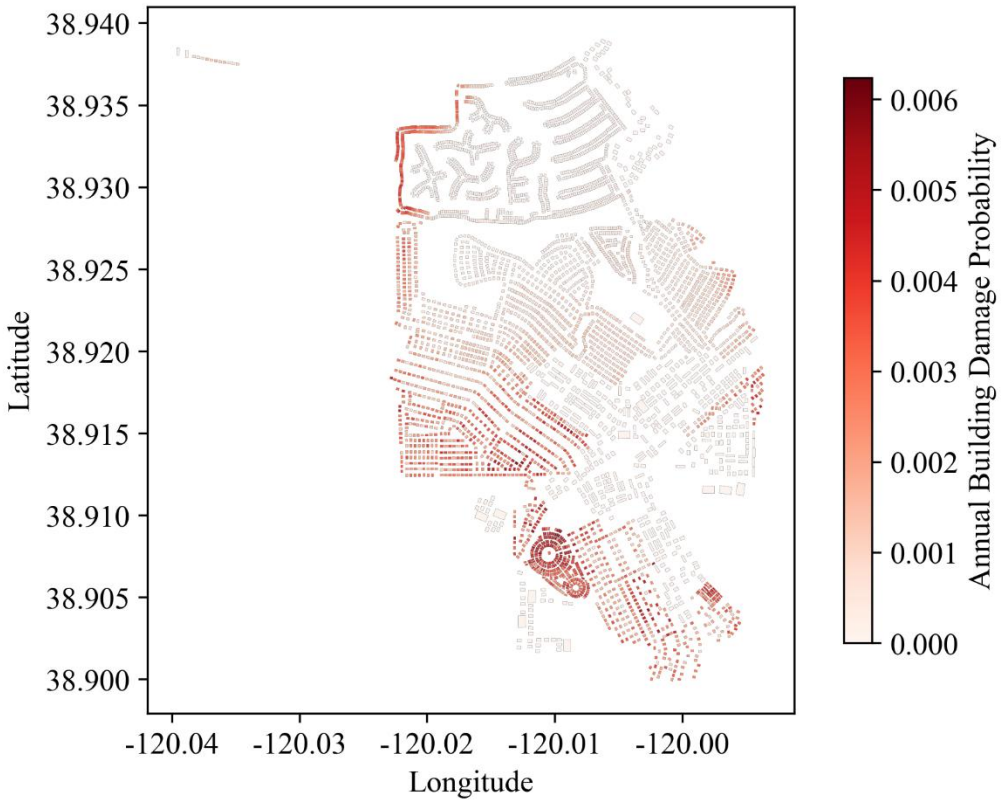

Figure 6 Annual building damage probability with consideration of burn probability of the town of forest under given weather and building attribute conditions.

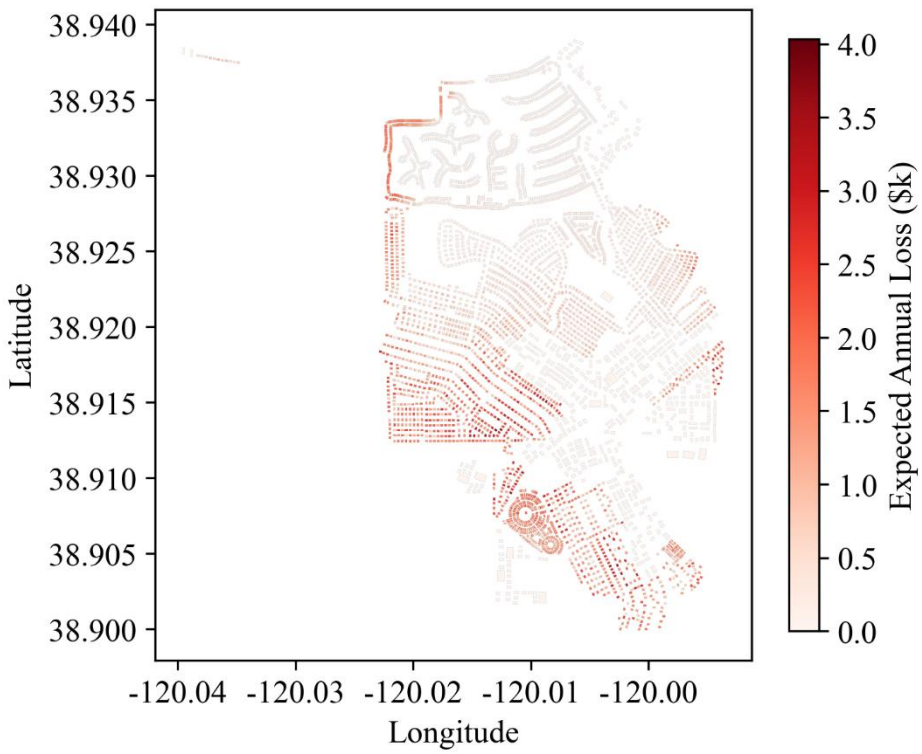

Figure 7 EAL of the town of forest under given weather and building attribute conditions.

4.4 Sensitivity Analyses

In addition to the predefined weather condition, total EAL of the town of forest are also estimated to characterize how community-scale wildfire consequence changes under different environmental conditions. In this analysis, wind speed and 10-hour fuel moisture content, denoted as MC10, are selected as hazard-intensity variables. For each combination of wind speed and MC10, the building damage model is re-evaluated, and the corresponding total EALs are calculated. Figure 8 presents the corresponding surface for total EAL. The results indicate that wind speed and MC10 have a strong influence on predicted wildfire consequences. Across the tested parameter space, the total EAL for the town of forest varies approximately from $46 thousand to $5,001 thousand.

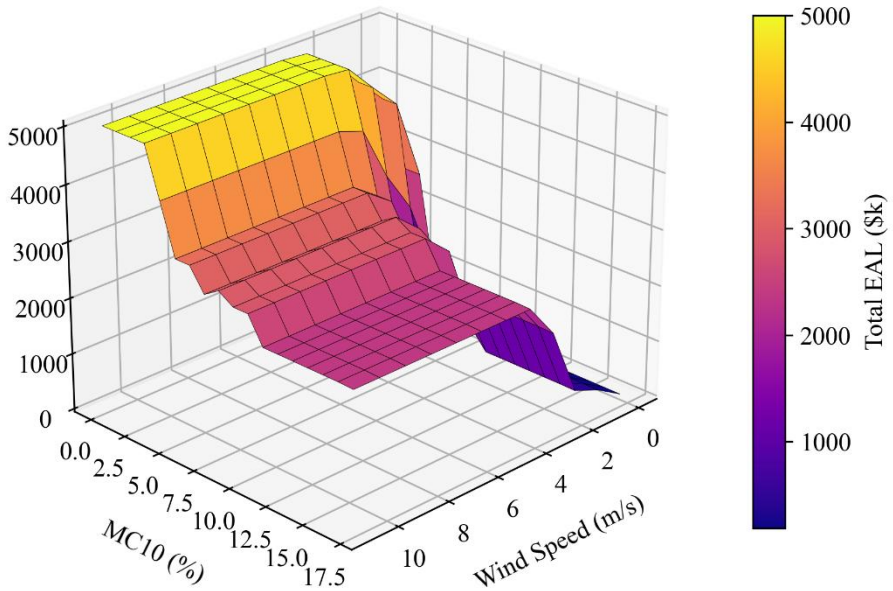

Figure 8 Fragility surfaces showing (a) predicted number of damaged buildings and (b) total expected loss under different wind speed and MC10 conditions.

5. Downstream Use of Model

The most immediate downstream use of this workflow is community-scale WUI risk assessment. Because the outputs quantify both the burn probability and the building damage probability, the model can help identify high-risk neighborhoods, especially in areas where vegetation, built density, and vulnerable building features combined to elevate consequence. This is directly aligned with the exercise's emphasis on mitigation, planning, and risk-and-resilience assessment.

A second downstream use is mitigation prioritization. Since the model includes interpretable building and exposure features, it can be used to test the likely benefit of interventions such as reducing near-structure vegetation, improving structure spacing, upgrading vents and eaves, or reducing combustible roof components.

A third use is resource allocation and pre-incident planning. Numerical maps of burn probability, building damage probability, and EAL can help emergency managers, planners, insurers, and local governments identify where mitigation funds or preparedness resources may have the highest benefit. Because the workflow produces community-wide consequence summaries in addition to building-level results, it can support both parcel-scale and neighborhood-scale decision making.

A fourth use is scenario testing under changing environmental conditions. The results shown for wind and MC10 demonstrate that the framework can be used to explore how different weather and fuel-moisture states affect outcomes. This supports stress testing, readiness assessment, and "what-if" analysis, which are valuable for both planning and operational awareness.

6. Assumptions, Parameters, and Validation

6.1 Key Assumptions

For the burn probability model, wildfire ignition is modeled as a spatially heterogeneous stochastic process. Ignition occurrence is represented using a Poisson point-process framework, where the expected ignition rate varies spatially based on historical fire occurrence information. This assumes that ignition events are conditionally independent given the local ignition rate.

In each fire spread simulation, fire spread is assumed to be deterministic once a specific realization of ignition location, wind speed, wind direction, fuel moisture, fuel condition, and landscape state is specified. These uncertainties are quantified using historic data. Landscape conditions, including fuel and topographic properties, are assumed to remain static during the simulation period. This simplification does not explicitly account for time-varying fuel moisture, suppression activities, dynamic fuel consumption, or changing fire-atmosphere interactions during the event. However, it provides a computationally efficient approach suitable for scenario-based community risk assessment. Finally, the effects of fire simulation modeling error are currently not currently considered in the process. Fire simulations are done based on FARSITE, which includes inherent sources of modeling error and uncertainty and lacks coupled fire-atmosphere interactions.

For the building damage model, the pre-trained machine-learning model is assumed to be transferable from the empirical training context to the synthetic Town of Forests scenario. The model was trained using DINS-based building damage records. Therefore, an important limitation is that the training data reflect California-specific fire events, inspection practices, building characteristics, and environmental conditions. Generalization into a synthetic community or to other geographic regions may introduce uncertainty.

The model also assumes that flame, radiation, and ember exposure can be represented through physics-guided proxy variables rather than direct high-fidelity combustion, heat-transfer, or ember-transport simulations at each structure. These proxies are designed to preserve physical interpretability while maintaining computational efficiency. However, they simplify complex WUI fire processes and may not capture all details of structure-to-structure spread, ember accumulation, localized shielding, or fine-scale building-envelope failure.

Furthermore, in the absence of detailed, building-specific information, commercial and industrial structures are assumed to be non-burnable and, therefore, not susceptible to ignition or damage from wildfire exposure. As a result, these structures are excluded from the building damage and expected annual loss (EAL) calculations. The analysis is restricted to residential buildings, which are assumed to be burnable and for which damage probabilities and EAL metrics are explicitly evaluated.

6.2 Key Parameter Choices

Several important parameter choices are made in this exercise. The EAL is calculated by combining the burn probability, building damage probability, with the building area and an assumed unit loss value. In this exercise, a unit loss of $200 per square foot was used. This assumption provides a simple and transparent way to estimate community-scale economic

loss, but it does not represent detailed replacement cost, contents loss, indirect losses, or socioeconomic consequences. Feature rephrasing is conducted to improve the ability of generalization of the model. More details of the rephrase process can be found in the code attached, and in the upcoming publication.

6.3 Validation

The BP model has been validated using historic observations for a few study sites across California. Details are included in our under-review manuscript and can be shared upon request.

The building-damage model was developed from wildfire damage observations and trained using DINS-based structure damage data. The broader model development process included evaluation against observed building damage outcomes and interpretation of feature importance using model explainability methods. The overall accuracy of the ML model is 0.76 and most of the features have reasonable trends and impact.

6.4 Limitations

Several limitations should be noted. First, the burn probability model depends on the assumed distributions of ignition, wind, and fuel-moisture conditions. If these distributions do not represent the actual range of possible conditions, the resulting burn probability estimates may be biased.

Second, the machine-learning model was trained using observed wildfire damage data from a specific empirical context. Although the model includes physics-guided features to improve interpretability and transferability, its predictions may still be affected by differences between the training data and the synthetic Town of Forests scenario.

Third, the wind-speed variable used in model training is based on regional-scale ERA5 data, whereas local wind conditions in WUI fire environments can vary substantially due to terrain, vegetation, and local atmospheric effects. This scale difference may introduce uncertainty when translating between regional weather descriptors and local exposure conditions.

Finally, the current framework does not explicitly simulate detailed structure-to-structure fire progression, suppression actions, evacuation behavior, or time-dependent damage accumulation. Instead, these processes are represented indirectly through exposure proxies and probabilistic damage prediction. Therefore, the results should be interpreted as scenario-based risk estimates rather than deterministic predictions of a specific fire event.